

Q1 2021 (31 Aug Shift 2)

Let B be the centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$.

Let the tangents at two points P and Q on the circle intersect at the point $A(3, 1)$. Then $8 \cdot \left(\frac{\text{area } \triangle APQ}{\text{area } \triangle BPQ} \right)$ is equal to .

Q2 2021 (31 Aug Shift 2)

An angle of intersection of the curves, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

and $x^2 + y^2 = ab, a > b$, is :

(1) $\tan^{-1} \left(\frac{a+b}{\sqrt{ab}} \right)$

(2) $\tan^{-1} \left(\frac{a-b}{2\sqrt{ab}} \right)$

(3) $\tan^{-1} \left(\frac{a-b}{\sqrt{ab}} \right)$

(4) $\tan^{-1}(2\sqrt{ab})$

Q3 2021 (31 Aug Shift 1)

If the variable line $3x + 4y = \alpha$ lies between the two circles $(x - 1)^2 + (y - 1)^2 = 1$ and

$(x - 9)^2 + (y - 1)^2 = 4$ without intercepting a chord on either circle, then the sum of all the integral values of α is .

Q4 2021 (27 Aug Shift 2)

Two circles each of radius 5 units touch each other at the point $(1, 2)$. If the equation of their common tangent is $4x + 3y = 10$, and $C_1(\alpha, \beta)$ and $C_2(\gamma, \delta)$, $C_1 \neq C_2$ are their centres, then $|(\alpha + \beta)(\gamma + \delta)|$ is equal to .

Q5 2021 (27 Aug Shift 1)

Let the equation $x^2 + y^2 + px + (1 - p)y + 5 = 0$ represent circles of varying radius $r \in (0, 5]$. Then the number of elements in the set $S = \{q : q = p^2 \text{ and } q \text{ is an integer}\}$ is .

Q6 2021 (26 Aug Shift 2)

A circle C touches the line $x = 2y$ at the point $(2, 1)$ and intersects the circle $C_1 : x^2 + y^2 + 2y - 5 = 0$ at two points P and Q such that PQ is a diameter of C_1 . Then the diameter of C is :

(1) $7\sqrt{5}$

(2) 15

(3) $\sqrt{285}$

(4) $4\sqrt{15}$

Q7 2021 (26 Aug Shift 1)

The locus of a point, which moves such that the sum of squares of its distances from the points $(0, 0), (1, 0), (0, 1), (1, 1)$ is 18 units, is a circle of diameter d . Then d^2 is equal to .

Q8 2021 (26 Aug Shift 1)

If a line along a chord of the circle $4x^2 + 4y^2 + 120x + 675 = 0$, passes through the point $(-30, 0)$ and is tangent to the parabola $y^2 = 30x$, then the length of this chord is :

(1) 5

(2) 7

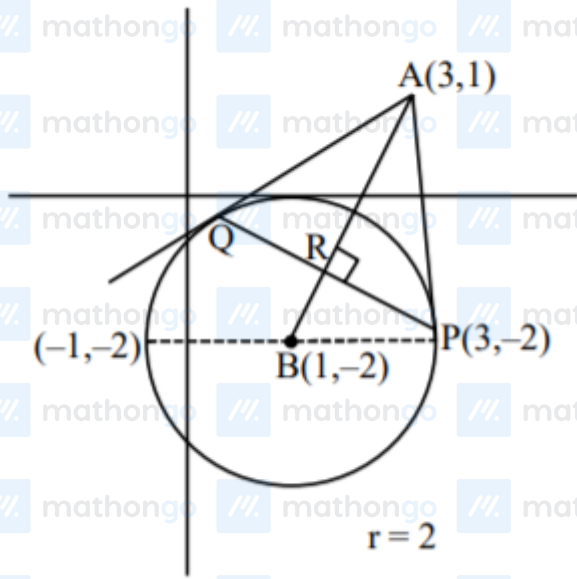
(3) $5\sqrt{3}$

(4) $3\sqrt{5}$

Answer Key

Q1 (18)**Q2 (3)****Q3 (165)****Q4 (40)****Q5 (61)****Q6 (1)****Q7 (16)****Q8 (4)**

Q1 (18)



$$\tan \theta = \frac{3}{2}$$

$$\frac{\text{Area } \triangle APQ}{\text{Area } \triangle BPQ} = \frac{AR}{RB} = \frac{3 \sin \theta}{2 \cos \theta} = \frac{9}{4}$$

$$8 \left(\frac{\text{Area } \triangle APQ}{\text{Area } \triangle BPQ} \right) = 18$$

Q2 (3)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, x^2 + y^2 = ab$$

$$\frac{2x_1}{a^2} + \frac{2y_1 y'}{b^2} = 0$$

$$\Rightarrow y'_1 = \frac{-x_1}{a^2} \cdot \frac{b^2}{y_1} \dots (1)$$

$$\therefore 2x_1 + 2y_1 y' = 0$$

$$\Rightarrow y'_2 = \frac{-x_1}{y_1} \dots (2)$$

Here (x_1, y_1) is point of intersection of both curves

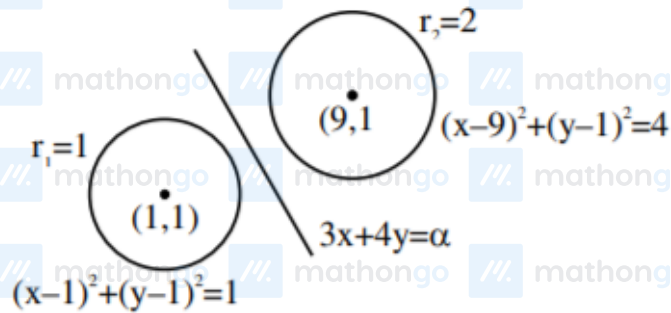
$$\therefore x_1^2 = \frac{a^2 b}{a+b}, y_1^2 = \frac{ab^2}{a+b}$$

$$\therefore \tan \theta = \left| \frac{y'_1 - y'_2}{1 + y'_1 y'_2} \right| = \left| \frac{\frac{-x_1 b^2}{a^2 y_1} + \frac{x_1}{y_1}}{1 + \frac{x_1^2 b^2}{a^2 y_1^2}} \right|$$

$$\tan \theta = \left| \frac{-b^2 x_1 y_1 + a^2 x_1 y_1}{a^2 y_1^2 + b^2 x_1^2} \right|$$

$$\tan \theta = \left| \frac{a-b}{\sqrt{ab}} \right|$$

Q3 (165)



Both centres should lie on either side of the line as well as line can be tangent to circle.

$$(3 + 4 - \alpha) \cdot (27 + 4 - \alpha) < 0$$

$$(7 - \alpha) \cdot (31 - \alpha) < 0 \Rightarrow \alpha \in (7, 31) \dots (1)$$

d_1 = distance of $(1, 1)$ from line

d_2 = distance of $(9, 1)$ from line

$$d_1 \geq r_1 \Rightarrow \frac{|7-\alpha|}{5} \geq 1 \Rightarrow \alpha \in (-\infty, 2] \cup [12, \infty) \dots (2)$$

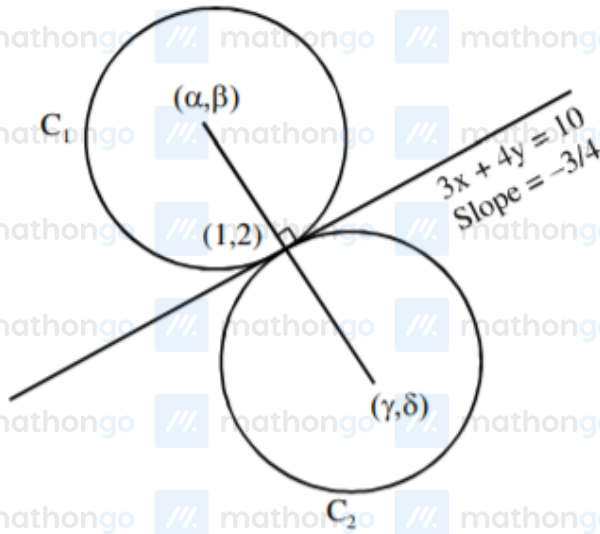
$$d_2 \geq r_2 \Rightarrow \frac{|31-\alpha|}{5} \geq 2 \Rightarrow \alpha \in (-\infty, 21] \cup [41, \infty) \dots (3)$$

$$(1) \cap (2) \cap (3) \Rightarrow \alpha \in [12, 21]$$

Sum of integers = 165

Q4 (40)

Slope of line joining centres of circles $= \frac{4}{3} = \tan \theta$



$$\Rightarrow \cos \theta = \frac{3}{5}, \sin \theta = \frac{4}{5}$$

Now using parametric form

$$\frac{x-1}{\cos \theta} = \frac{y-2}{\sin \theta} = \pm 5$$

$$\oplus (x, y) = (1 + 5 \cos \theta, 2 + 5 \sin \theta)$$

$$(\alpha, \beta) = (4, 6)$$

$$\ominus (x, y) = (\gamma, \delta) = (1 - 5 \cos \theta, 2 - 5 \sin \theta)$$

$$(\gamma, \delta) = (-2, -2)$$

$$\Rightarrow |(\alpha + \beta)(\gamma + \delta)| = |10x - 4| = 40$$

Q5 (61)

$$r = \sqrt{\frac{p^2}{4} + \frac{(1-p)^2}{4}} - 5 = \frac{\sqrt{2p^2 - 2p - 19}}{2}$$

Since, $r \in (0, 5]$

So, $0 < 2p^2 - 2p - 19 \leq 100$

$$\Rightarrow p \in \left[\frac{1-\sqrt{239}}{2}, \frac{1+\sqrt{39}}{2} \right) \cup \left(\frac{1+\sqrt{39}}{2}, \frac{1+\sqrt{239}}{2} \right] \text{ so, number of integral values of } p^2 \text{ is 61}$$

Q6 (1)

$$(x-2)^2 + (y-1)^2 + \lambda(x-2y) = 0$$

$$C : x^2 + y^2 + x(\lambda - 4) + y(-2 - 2\lambda) + 5 = 0$$

$$C_1 : x^2 + y^2 + 2y - 5 = 0$$

$$S_1 - S_2 = 0 \text{ (Equation of PQ)}$$

$$(\lambda - 4)x - (2\lambda + 4)y + 10 = 0 \text{ Passes through } (0, -1)$$

$$\Rightarrow \lambda = -7$$

$$C : x^2 + y^2 - 11x + 12y + 5 = 0$$

$$= \frac{\sqrt{245}}{4}$$

$$\text{Diameter} = 7\sqrt{5}$$

Q7 (16)

Let $P(x, y)$

$$x^2 + y^2 + x^2 + (y - 1)^2 + (x - 1)^2 + y^2 + (x - 1)^2 + (y - 1)^2;$$

$$\Rightarrow 4(x^2 + y^2) - 4y - 4x = 14$$

$$\Rightarrow x^2 + y^2 - x - y - \frac{7}{2} = 0$$

$$d = 2\sqrt{\frac{1}{4} + \frac{1}{4} + \frac{7}{2}}$$

$$\Rightarrow d^2 = 16$$

Q8 (4)

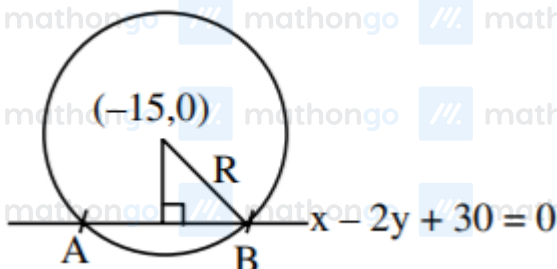
Equation of tangent to $y^2 = 30x$

$$y = mx + \frac{30}{4m}$$

$$\text{Pass thru } (-30, 0) : a = -30m + \frac{30}{4m} \Rightarrow m^2 = 1/4$$

$$\Rightarrow m = \frac{1}{2} \text{ or } m = -\frac{1}{2}$$

$$\text{At } m = \frac{1}{2} : y = \frac{x}{2} + 15 \Rightarrow x - 2y + 30 = 0$$



$$P = \frac{15}{\sqrt{5}}$$

$$\ell_{AB} = 2\sqrt{R^2 - P^2} = 2\sqrt{\frac{225}{4} - \frac{225}{5}}$$
$$\Rightarrow \ell_{AB} = 30 \cdot \sqrt{\frac{1}{20}} = \frac{15}{\sqrt{5}} = 3\sqrt{5}$$